1

(1) $\lambda = 2$

eigenvalue of $\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$?

$(A - \lambda \bar{I}) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\det(A - \lambda \bar{I}) = 0$

$A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$

$\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

$\det\begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} = 6 - 6 = 0$

lin indep, has non-trivialsolution

so 2 is eigenvalue

(3) $\bar{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ an eigenvector of $A = \begin{bmatrix} -3 & 1 \\ -3 & 8 \end{bmatrix}$?

eigenvalue?

2

$A\bar{u} = \lambda \bar{u}$

$\begin{bmatrix} -3 & 1 \\ -3 & 8 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$\begin{bmatrix} -3 + 4 \\ -3 + 32 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 29 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

not eigenvector

(5) $\bar{u} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ $A = \begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$ is $A\bar{u} = \lambda\bar{u}$? if so, find λ.

$\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 + (-21) + 9 \\ -16 + 15 + 1 \\ 8 - 12 + 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$

eigenvalue $= 0 = \lambda$

s

3

(7) $\lambda = 4 \quad A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$

is $A\bar{u} = \lambda \bar{u}$? if so, find $\bar{x}$

$\det(A - \bar{u}I) = 0$

$$\det\left(\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}\right) = \begin{vmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{vmatrix} =$$

$$= \begin{vmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 4 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$ lin. indep

$A\bar{u} = \lambda\bar{u}$ has a non-trivial sol-n, λ is an eigenvalue, as $(A - \lambda I)\cdot\bar{x} = \bar{0}$.

$$\left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 2 & -1 & 1 & 0 \\ -3 & 4 & 1 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\begin{cases} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \quad x_1 = x_2 = -x_3$

~~eigenvector =~~

eigenvector $= \bar{u} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

4

(8) Basis for eigenspace (8 - 15)

$A = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix} \quad \lambda = 1, 5$

$\det(A - \bar{u}I) = 0$

~~$\begin{vmatrix} (5-1,5) & 0 \\ 2 & (1-1,5) \end{vmatrix} = 3,5 \cdot (-0,5) - 2 =$~~

$\lambda = 1)$

$\begin{vmatrix} (5-1) & 0 \\ 2 & (1-1) \end{vmatrix} = 0$

as $(A - \lambda I)\cdot\bar{x} = 0$

$\left[\begin{array}{cc|c} 4 & 0 & 0 \\ 2 & 0 & 0 \end{array}\right] \quad \begin{array}{l} 4x_1 = 0 \\ 2x_1 = 0 \end{array} \quad x_1 = 0$

$x_2 = $ free

e/space = span $\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ Basis $= \left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$

5

$\lambda = 5;$ $|A - \bar{u}\bar{I}|$

$$\begin{vmatrix} 5-5 & 0 \\ 2 & 1-5 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 2 & -4 \end{vmatrix} = 0 \quad \text{lin. indep}$$

$(A - \lambda \bar{I})\bar{x} = 0$

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 2 & -4 & 0 \end{array}\right] \quad \begin{array}{l} 2x_1 - 4x_2 = 0 \\ x_1 - 2x_2 = 0 \\ x_1 = 2x_2 \\ x_2 \text{ is free} \end{array}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

~~e/space~~ Basis $= \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ of e/space

e/space $= \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$

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$$A = \begin{bmatrix} 4 & -2 \\ -3 & 8 \end{bmatrix} \quad \lambda = 10$$

$|A - \lambda \bar{I}|$:

$$\det\left(\begin{bmatrix} 4-10 & -2 \\ -3 & 8-10 \end{bmatrix} \right) = \cancel{6 \cdot 80 - 6} = 6 - 6 = 0$$

lin. indep

$(A - \lambda I)\bar{x} = 0$

$$\begin{bmatrix} -6 & -2 \\ -3 & -1 \end{bmatrix} \bar{x} = 0 \quad \left[\begin{array}{cc|c} 3 & 1 & 0 \\ 3 & 1 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 1 & \frac{1}{3} & 0 \end{array}\right] \Rightarrow$$

6

$$\Rightarrow \left[\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{array}\right] \therefore \quad \begin{array}{l} x_1 + \frac{x_2}{3} = 0 \\ x_1 = -\frac{x_2}{3} \\ x_2 \text{ is free} \end{array}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{x_2}{3} \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix} \Rightarrow \text{Basis} = \left\{ \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix} \right\}$$

e/space $= \text{span} \left\{ \begin{bmatrix} -\frac{1}{3} \\ 1 \end{bmatrix} \right\}$

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$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad \lambda = 1, 2, 3$$

$\lambda = 1$; $|A - \lambda I| = 0$

$$\det\left(\begin{bmatrix} (4-1) & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \right) = 0$$

7

$(A-\lambda I)\bar{x}=0$

$$\begin{bmatrix}3&0&1\\-2&0&0\\-2&0&0\end{bmatrix}=0 \quad \left[\begin{array}{ccc|c}3&0&1&0\\2&0&0&0\\2&0&0&0\end{array}\right] \Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{3}&0\\2&0&0&0\\2&0&0&0\end{array}\right] \Rightarrow$$

$$\cancel{x_1+\frac{x_3}{3}=0} \quad \cancel{2x_1=0} \quad \cancel{2x} \quad \cancel{x_1+\frac{x_3}{3}=0} \quad \cancel{-x_3\frac{2}{3}=0}$$

$$\Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{3}&0\\0&0&-\frac{2}{3}&0\\0&0&-\frac{2}{3}&0\end{array}\right] \Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{3}&0\\0&0&2&0\\0&0&2&0\end{array}\right] \Rightarrow$$

$$\Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{3}&0\\0&0&1&0\\0&0&1&0\end{array}\right] \Rightarrow \left[\begin{array}{ccc|c}1&0&0&0\\0&0&1&0\\0&0&0&0\end{array}\right]$$

$x_1=0$ $\quad x_2$ is free

$x_3=0$

Basis $=\left\{\begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$ $\quad$ e/space = span $\left\{\begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$

$\lambda=2$:

$|A-\lambda I|=0$

$$\begin{vmatrix}2&0&1\\-2&-1&0\\-2&0&-1\end{vmatrix}=(-1)^2\cdot 2\cdot\begin{vmatrix}-1&0\\0&-1\end{vmatrix}+(-1)^4\cdot\begin{vmatrix}-2&-1\\-2&0\end{vmatrix}=$$

8

$=2\cdot(1)-2=0$

$(A-\lambda I)\bar{x}=0$

$$\left[\begin{array}{ccc|c}2&0&1&0\\-2&-1&0&0\\-2&0&-1&0\end{array}\right] \Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{2}&0\\0&-1&1&0\\0&0&0&0\end{array}\right] \Rightarrow \cancel{\left[\begin{array}{ccc|c}1&0&0&0\\0&1&0&0\\0&0&1&0\end{array}\right]}$$

$\cancel{x_1=x_2=x_3=0}$

$$\Rightarrow \left[\begin{array}{ccc|c}1&0&\frac{1}{2}&0\\0&1&-1&0\\0&0&0&0\end{array}\right] \quad \begin{cases}x_1+\frac{x_3}{2}=0\\x_2-x_3=0\end{cases} \quad x_3 = \text{free}$$

$x_1=-\frac{x_3}{2}$ $\quad x_2=x_3$

Basis $=\left\{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right\}=\left\{\begin{bmatrix}-\frac{x_3}{2}\\x_3\\x_3\end{bmatrix}\right\}=\left\{\begin{bmatrix}-\frac{1}{2}\\1\\1\end{bmatrix}\right\}$

e/space = span $\left\{\begin{bmatrix}-\frac{1}{2}\\1\\1\end{bmatrix}\right\}$

9

$\lambda = 3$

$(A - \lambda I)\bar{x} = 0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -2 & -2 & 0 & 0 \\ -2 & 0 & -2 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{cases} x_1 + x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \quad \begin{cases} x_1 = -x_3 \\ x_2 = x_3 \end{cases} \quad x_3 \text{ is free}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Basis $= \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$ e/space = span $\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

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$$A = \begin{bmatrix} 4 & 2 & 3 \\ -1 & 1 & -3 \\ 2 & 4 & 9 \end{bmatrix} \quad \lambda = 3$$

$(A - \lambda I)\bar{x} = 0$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ -1 & -2 & -3 & 0 \\ 2 & 4 & 6 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

10

$x_1 + 2x_2 + 3x_3 = 0$

x_2, x_3 is free

$x_1 = -2\underset{s}{x_2} - 3\underset{t}{x_3}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2s - 3t \\ s \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Basis $= \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ e/space = span $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

17 $\lambda = ?$ $|A - \lambda I| = 0$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix} \quad \begin{vmatrix} -\lambda & 0 & 0 \\ 0 & 2-\lambda & 5 \\ 0 & 0 & -1-\lambda \end{vmatrix} = 0$$

$$-\lambda \begin{vmatrix} (2-\lambda) & 5 \\ 0 & (-1-\lambda) \end{vmatrix} = +\lambda(2-\lambda)(1+\lambda) =$$

11

$\lambda(2-\lambda)(1+\lambda) = \lambda(2+2\lambda-\lambda-\lambda^2) =$

$= -\lambda^3+\lambda^2+2\lambda = 0$

$\lambda^3-\lambda^2-2\lambda = 0$

$\lambda(\lambda^2-\lambda-2) = 0$

$\lambda = \frac{1 \pm \sqrt{1+4\cdot 2}}{2} = \begin{cases} \frac{1-3}{2} = -1 \\ \frac{1+3}{2} = 2 \end{cases}$

$\lambda = 0, 2, -1$

(13)

$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$

~~λ might be equal to 1, based~~

not inver~~t~~able, 0

(3)

$\begin{bmatrix} 3 & -2 \\ 1 & -8 \end{bmatrix}$

$A - \lambda \bar{I} = \begin{bmatrix} 3-\lambda & -2 \\ 1 & -1-\lambda \end{bmatrix}$

$\lambda = 1 \pm \sqrt{2}$

(11)

$\begin{bmatrix} 4 & 0 & 0 \\ 5 & 3 & 2 \\ -2 & 0 & 2 \end{bmatrix}$

$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 0 & 0 \\ 5 & 3-\lambda & 2 \\ -2 & 0 & 2-\lambda \end{vmatrix} = (4-\lambda)\begin{vmatrix} 3-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix} =$

$= (4-\lambda)(3-\lambda)(2-\lambda) = -\lambda^3+9\lambda^2-26\lambda+24$

(13) $\begin{bmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{bmatrix}$ $|A-\lambda \bar{I}| = \begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 9-\lambda & 0 \\ 5 & 8 & 3-\lambda \end{vmatrix} =$

$= 3-\lambda \begin{vmatrix} 6-\lambda & -2 \\ -2 & 9-\lambda \end{vmatrix} = -\lambda^3+18\lambda^2+95\lambda+150$

12

13

13
$$\begin{bmatrix} 4 & -1 & 0 & 2 \\ 0 & 3 & -4 & 6 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad |A-\lambda \underline{I}| = (4-\lambda)(3-\lambda)^2(1-\lambda)$$

4, 3, 3, 1 – eigenvalues

17
$$\begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ -5 & 1 & 0 & 0 & 0 \\ 3 & 8 & 0 & 0 & 0 \\ 0 & -7 & 2 & 1 & 0 \\ -4 & 1 & 9 & -2 & 3 \end{bmatrix}$$

$$|A-\lambda \underline{I}| = (3-\lambda)(1-\lambda)(-\lambda)(1-\lambda)(3-\lambda) =$$
$$= (3-\lambda)^2(1-\lambda)^2(-\lambda)$$

3, 3, 1, 1, 0 – eigenvalues